One request, three hops.

App → API Gateway → Backend → Database

THE ARCHITECTURE

Every call takes the same path.

01 · CLIENT

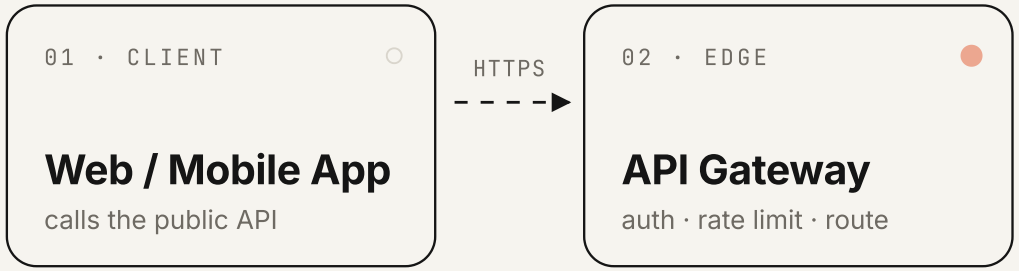
Web / Mobile App

calls the public API

THE ARCHITECTURE

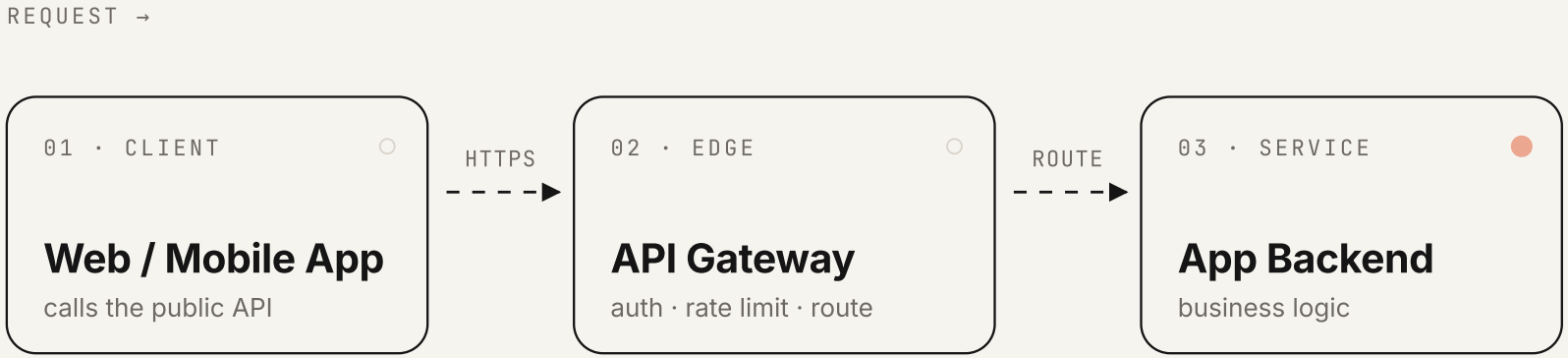
Every call takes the same path.

REQUEST →



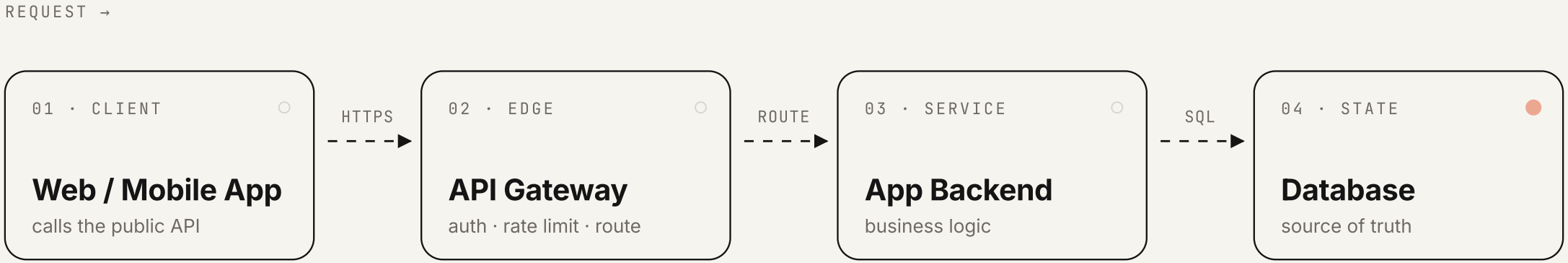
THE ARCHITECTURE

Every call takes the same path.



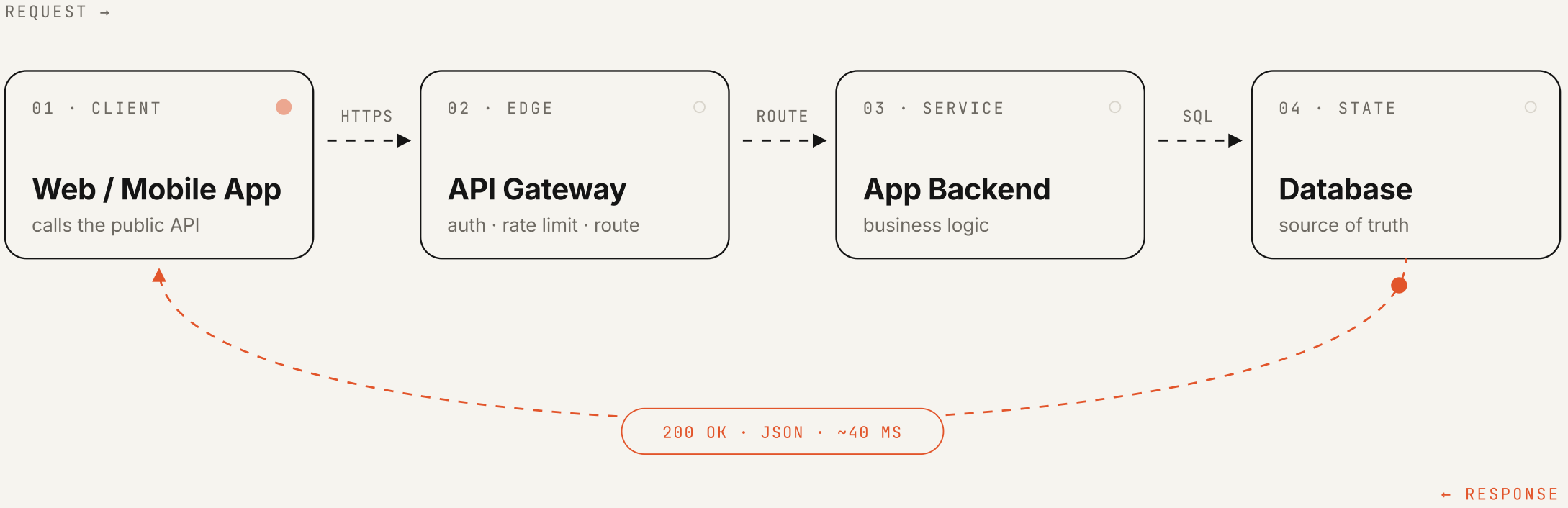
THE ARCHITECTURE

Every call takes the same path.

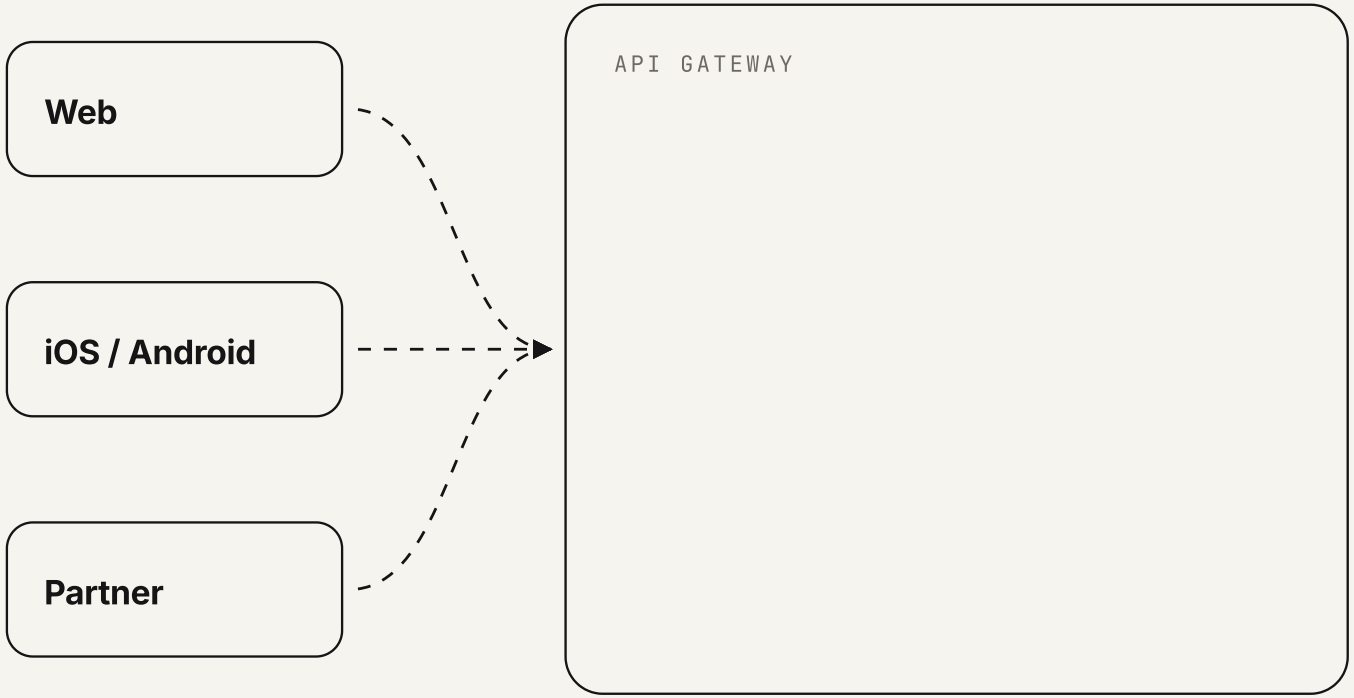


THE ARCHITECTURE

Every call takes the same path.

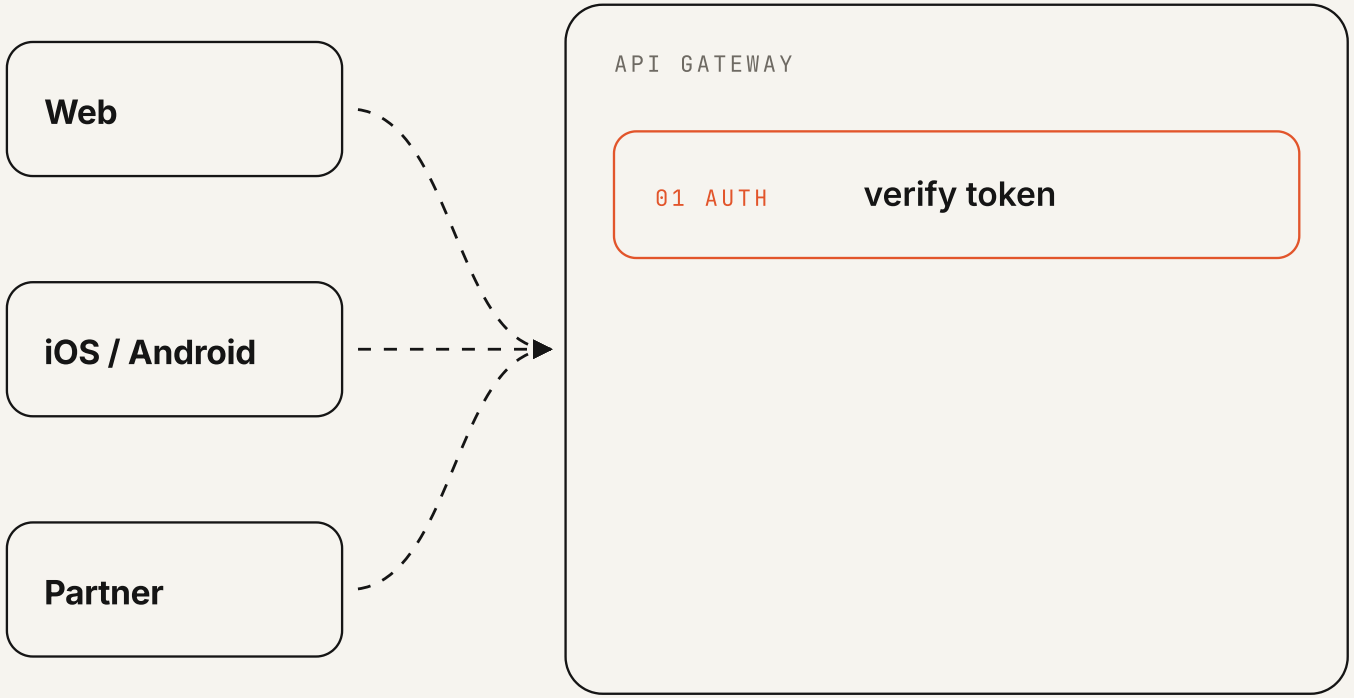


The front door says no early.



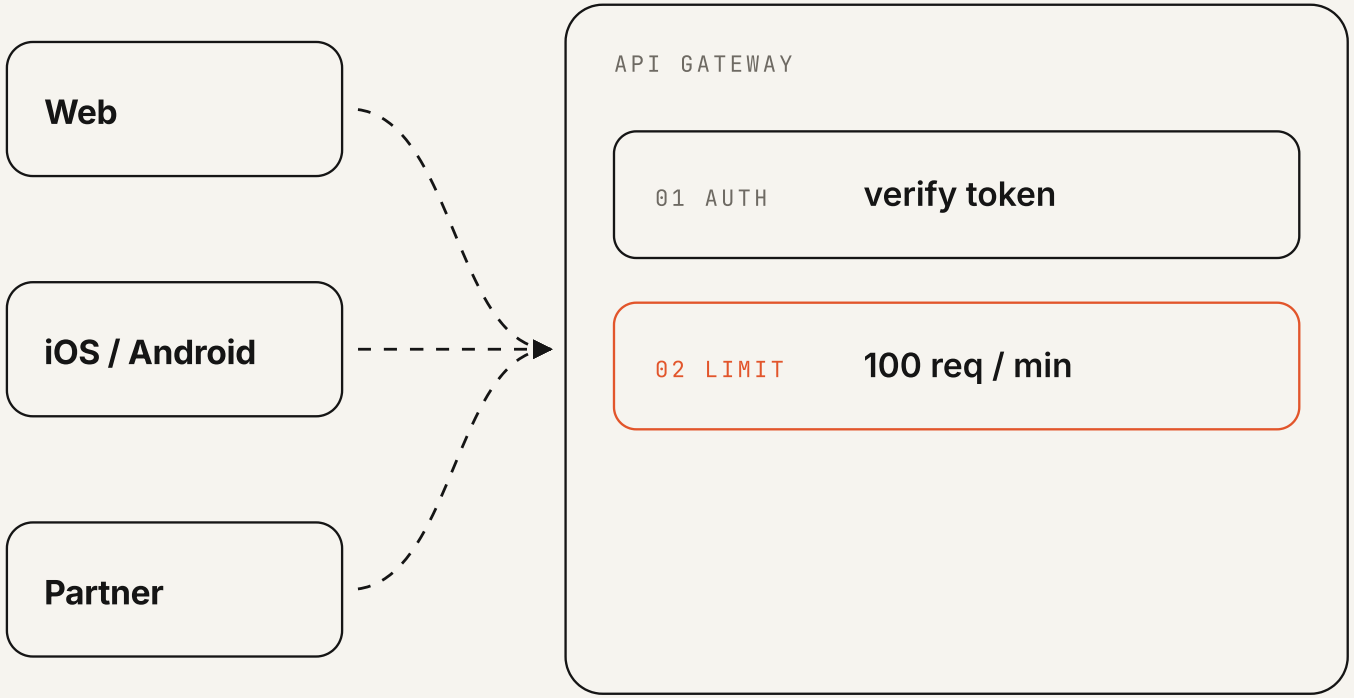
02 · API GATEWAY

The front door says no early.



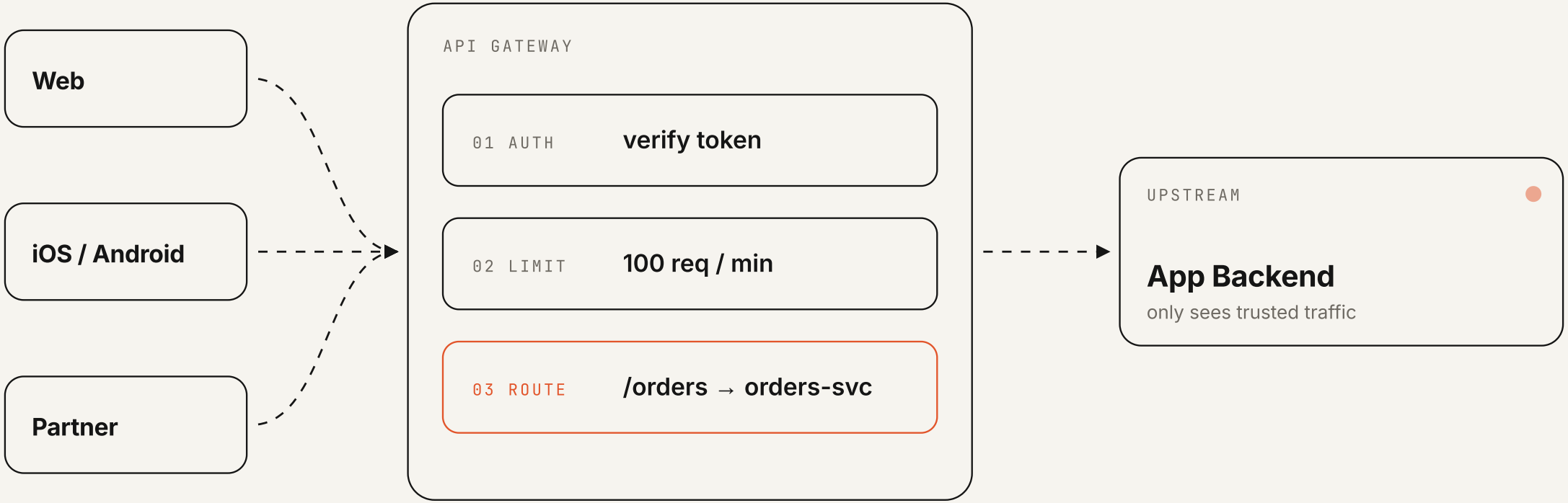
02 · API GATEWAY

The front door says no early.



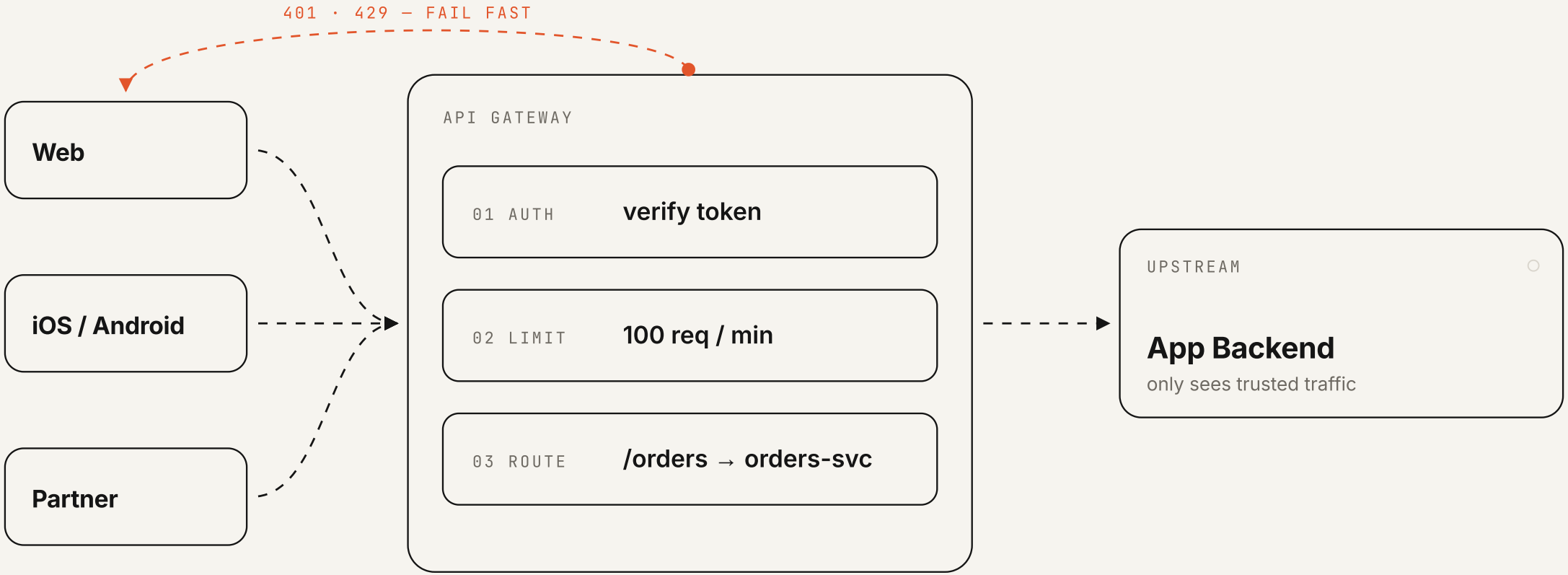
02 · API GATEWAY

The front door says no early.



02 · API GATEWAY

The front door says no early.



03 · APPLICATION BACKEND

Stateless, so it scales out.

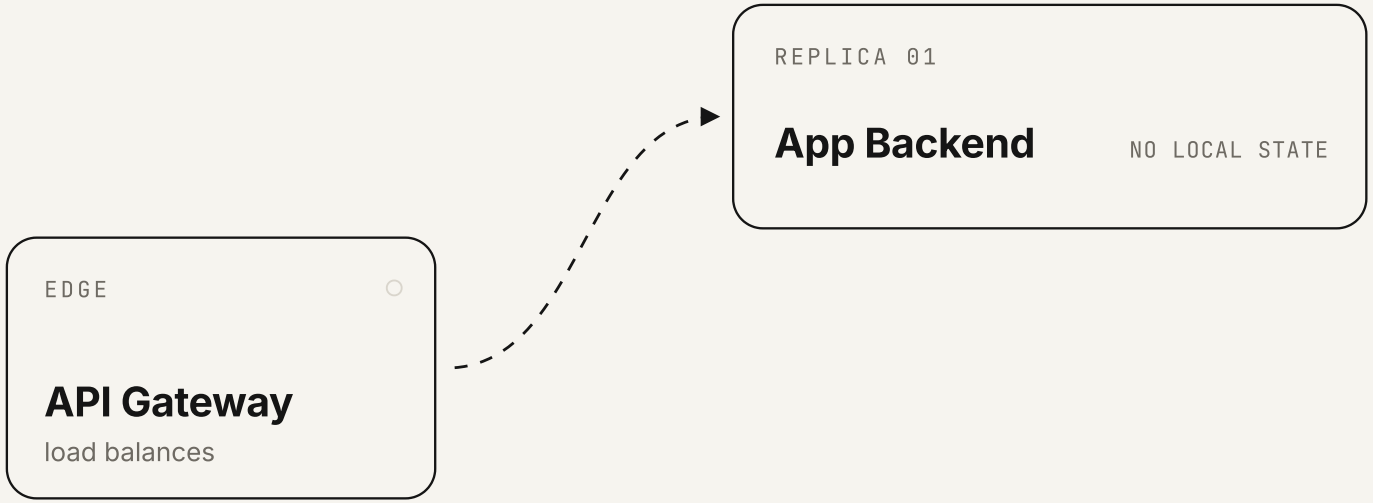
EDGE

API Gateway

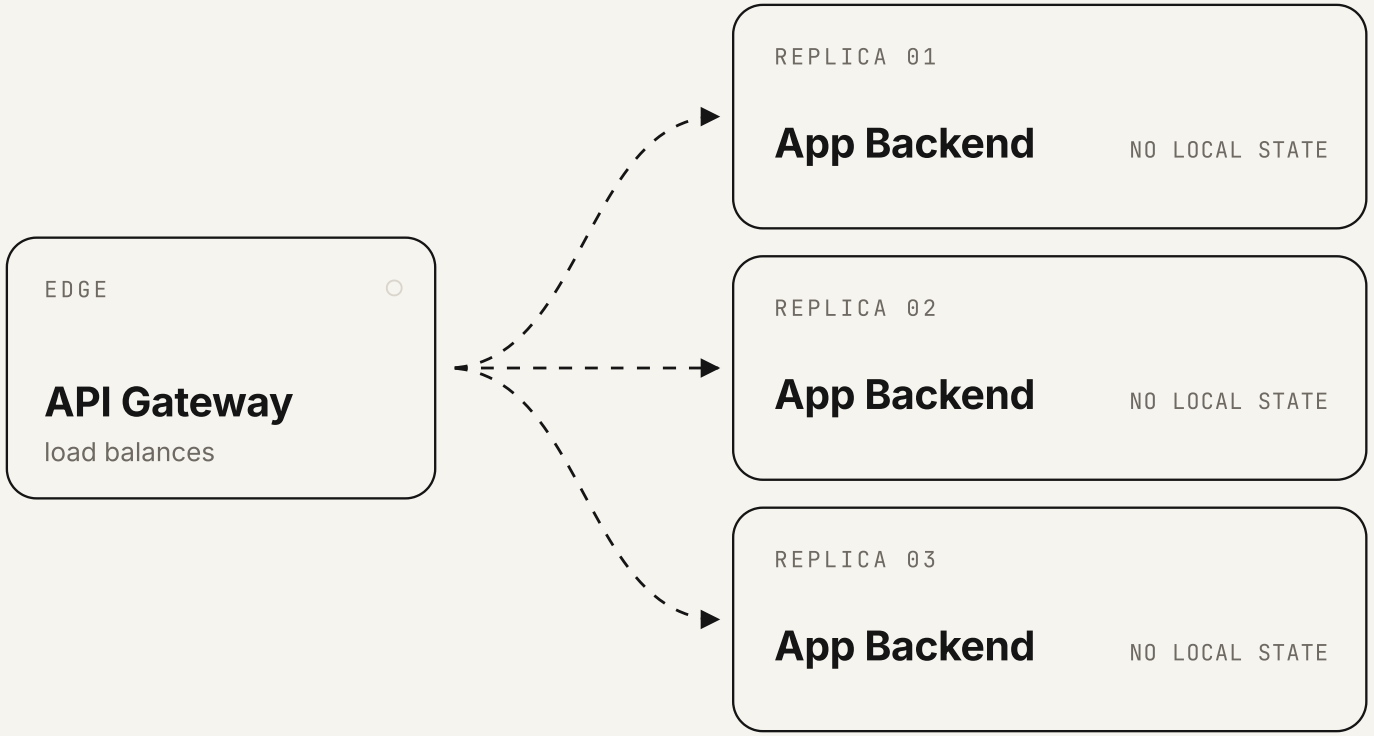
load balances

03 · APPLICATION BACKEND

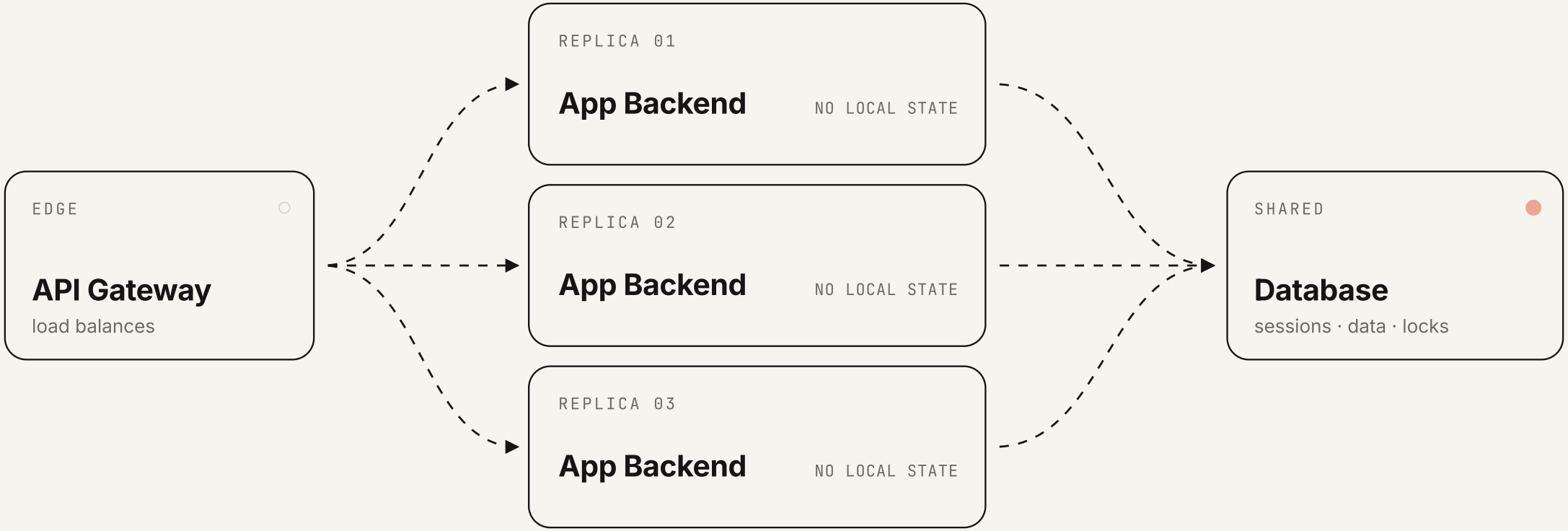
Stateless, so it scales out.



Stateless, so it scales out.



Stateless, so it scales out.



04 · DATABASE

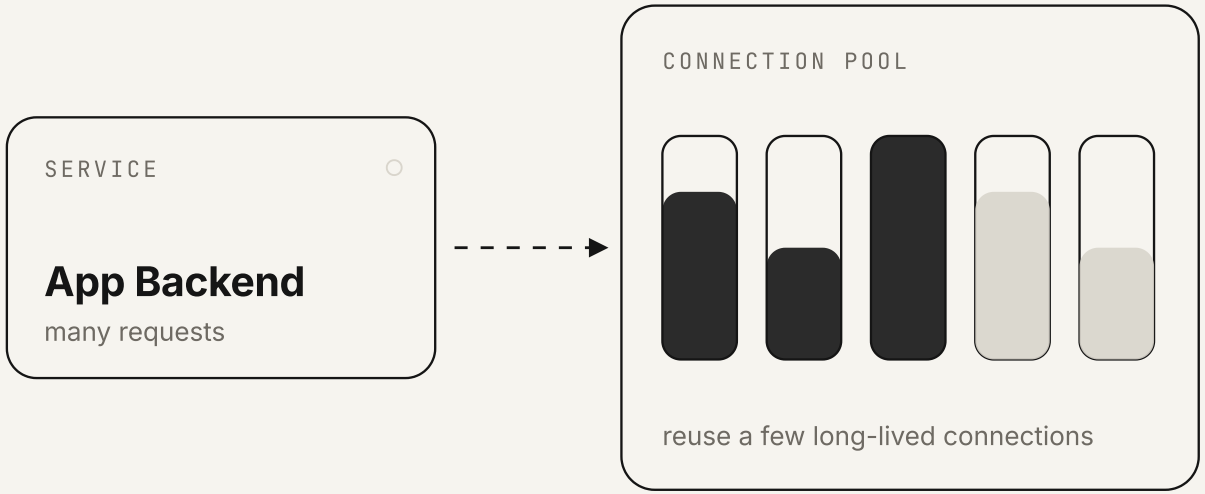
Writes go up, reads fan out.

SERVICE

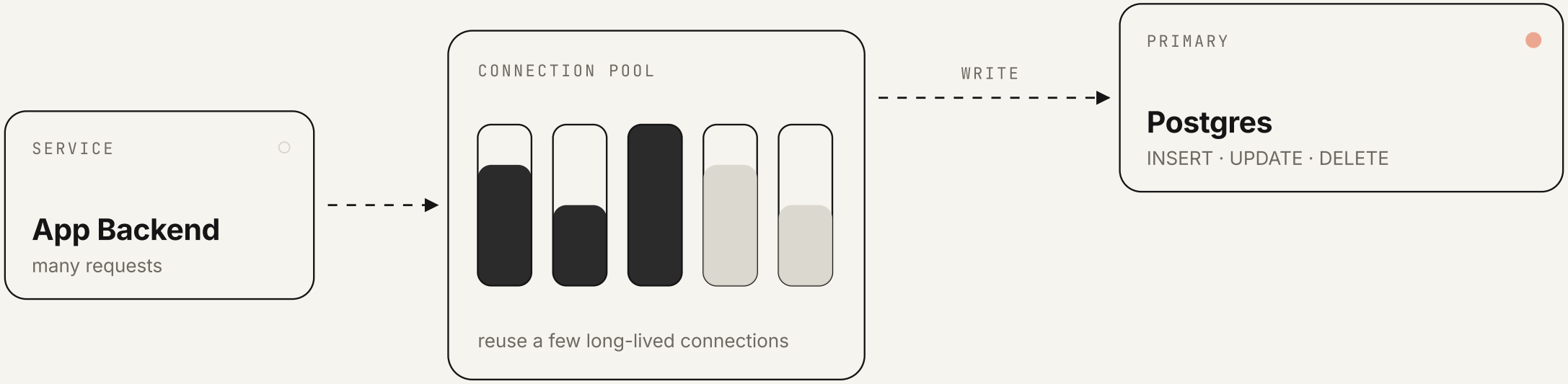
App Backend

many requests

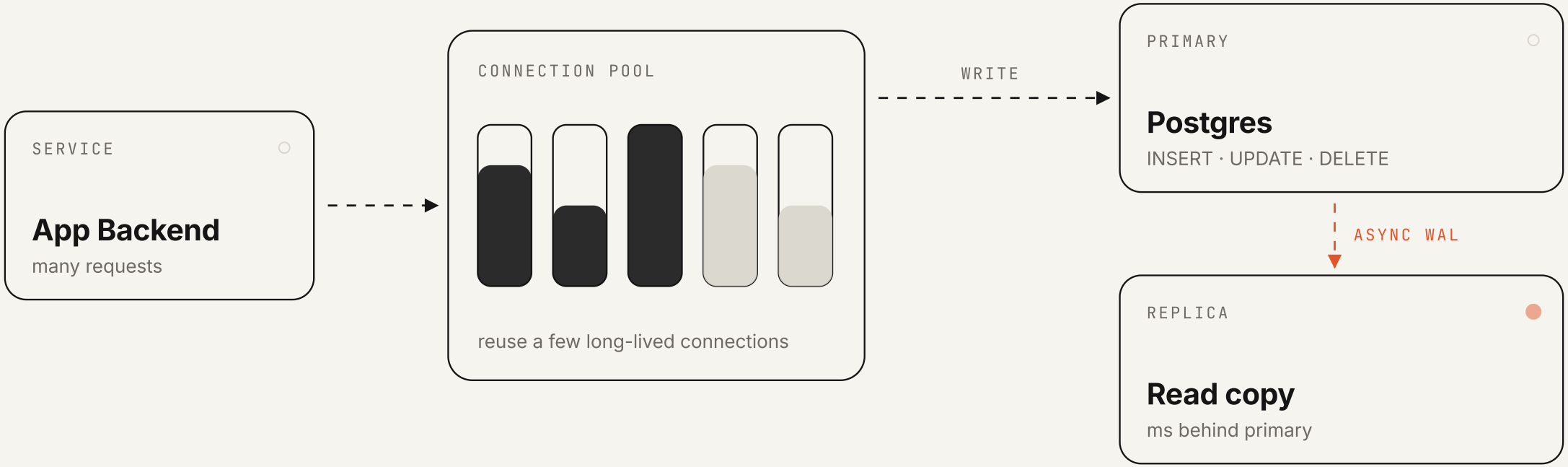
Writes go up, reads fan out.



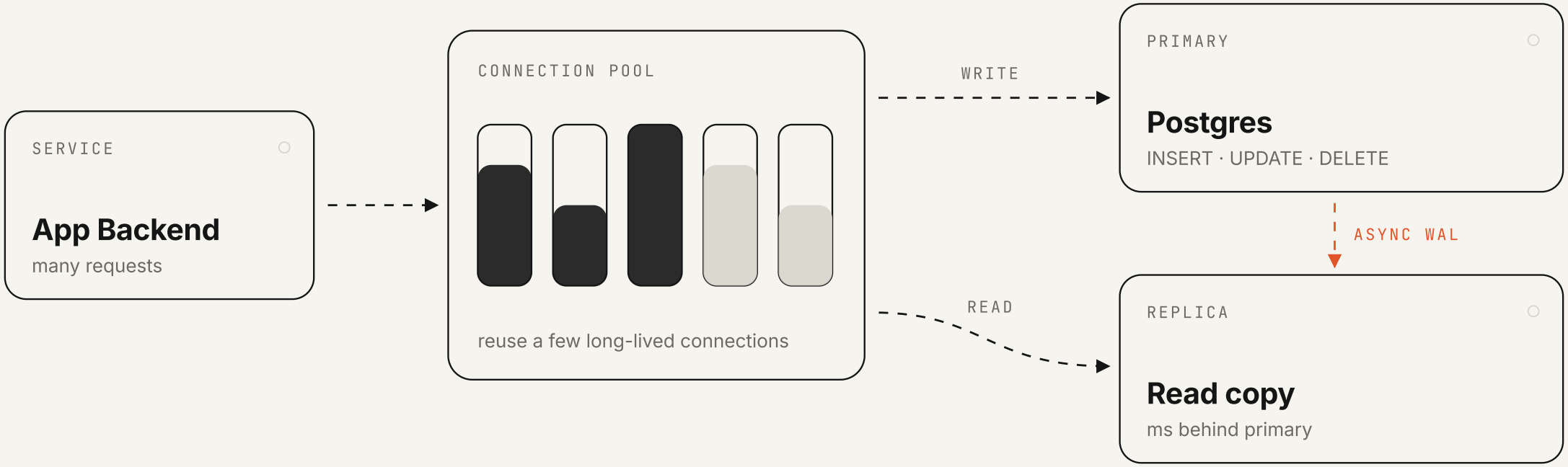
Writes go up, reads fan out.



Writes go up, reads fan out.

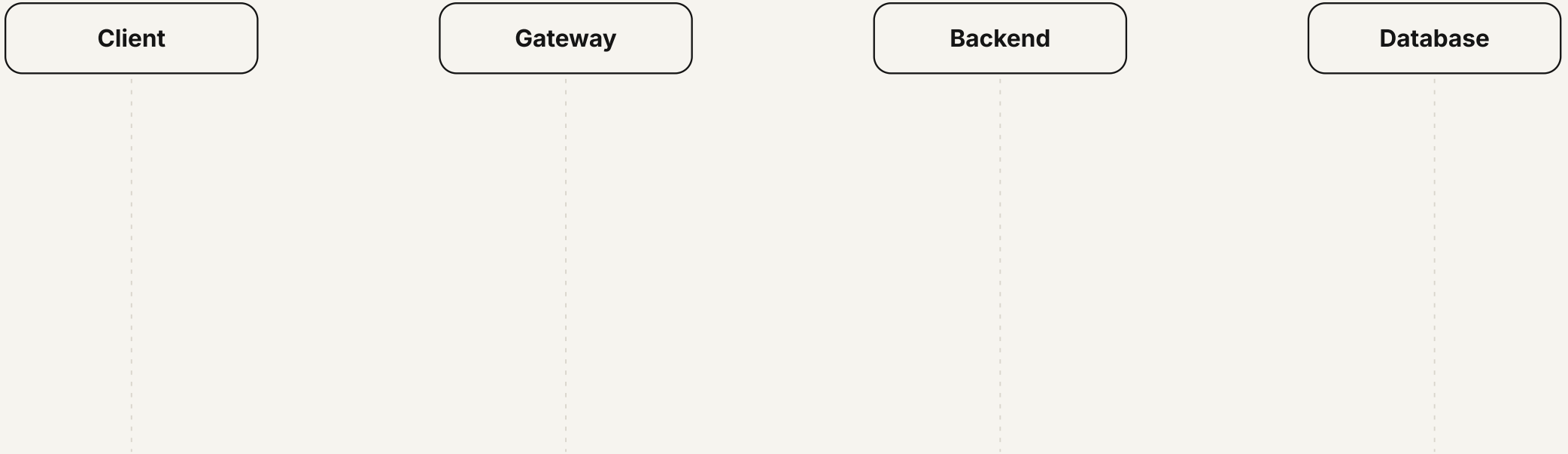


Writes go up, reads fan out.



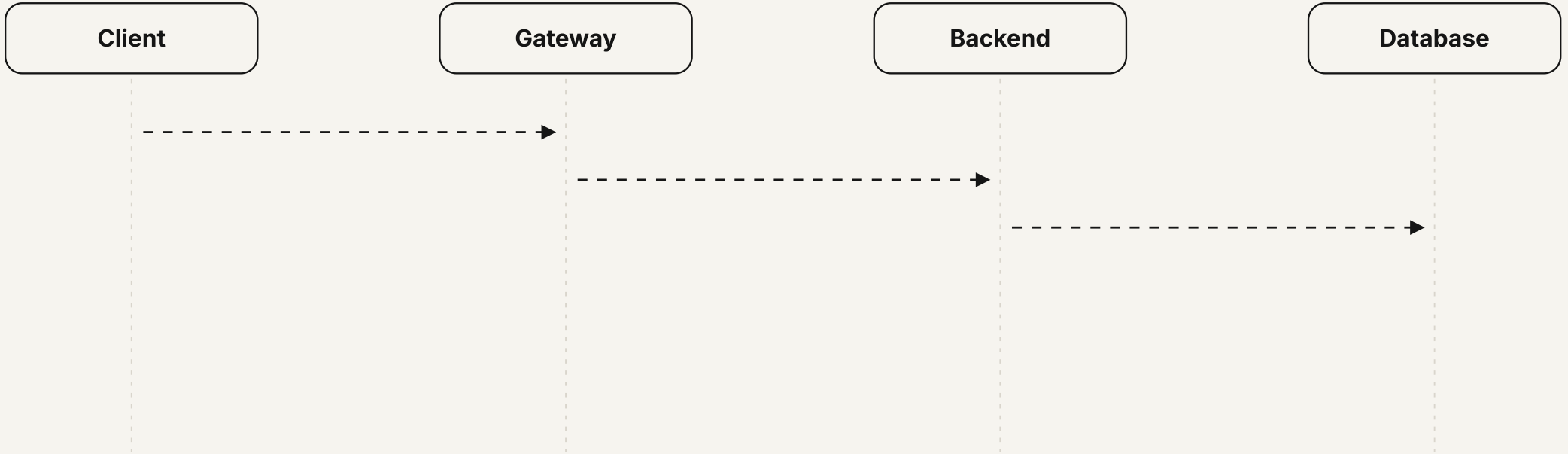
FEEDBACK LOOP

Every request owes a response.



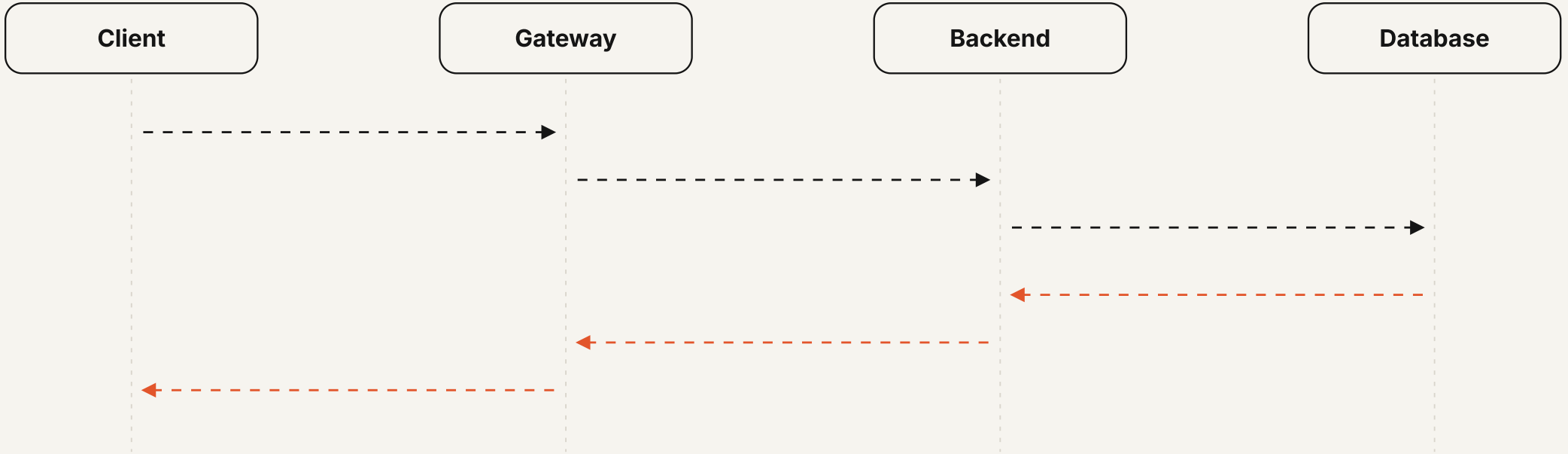
FEEDBACK LOOP

Every request owes a response.



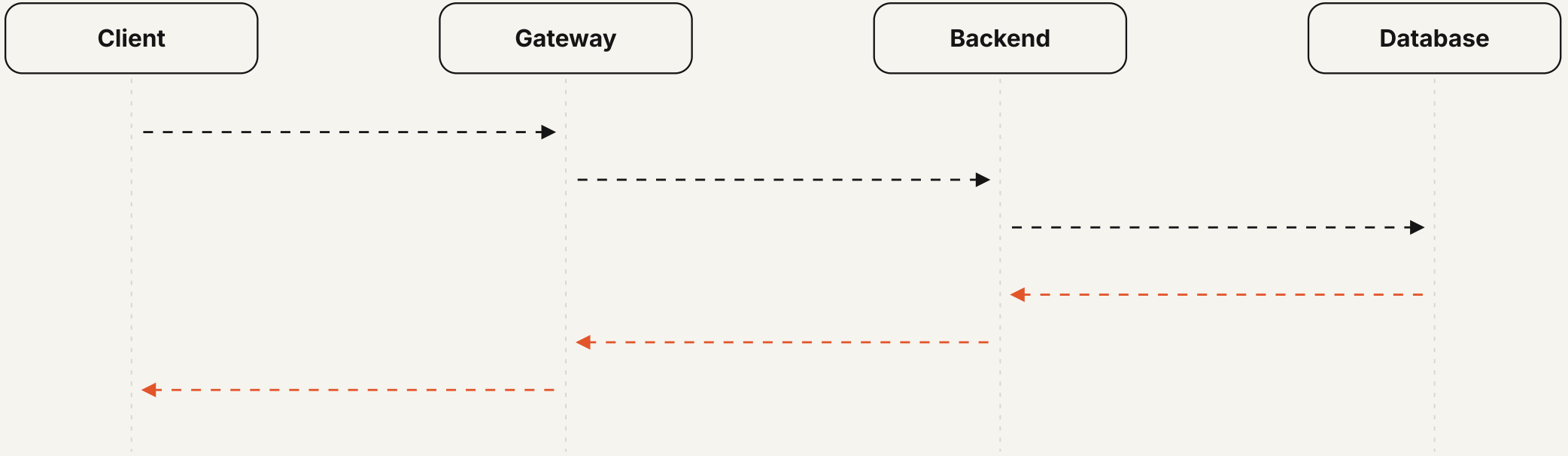
FEEDBACK LOOP

Every request owes a response.



FEEDBACK LOOP

Every request owes a response.



LATENCY BUDGET · 42 MS



TAKEAWAYS

Keep each hop boring.

TAKEAWAYS

Keep each hop boring.

02 · GATEWAY

Guard the edge

Reject bad traffic before it costs compute.

TAKEAWAYS

Keep each hop boring.

02 · GATEWAY

Guard the edge

Reject bad traffic before it costs compute.

03 · BACKEND

Hold no state

Any replica can serve any request.

TAKEAWAYS

Keep each hop boring.

02 · GATEWAY

Guard the edge

Reject bad traffic before it costs compute.

03 · BACKEND

Hold no state

Any replica can serve any request.

04 · DATABASE

Own the truth

Pool connections, split reads from writes.